MIL-K-81779C(AS)
11 July 1977
Superseding
MIL-K-81779B(AS)
6 December 1974

MILITARY SPECIFICATION

KIT, TOOL, SPECIAL
TK-222( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Special Tool Kit (STK) TK-222( )/AWM-23(V).

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue specified in AS-4053, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) shall be the applicable general specification for the STK described herein (see 6.6).

SPECIFICATION

Military

MIL-T-81734(AS)	Test Set, Airborne Weapon Control System, AN/AWM-23(V)

DRAWING

Hughes Aircraft Company

493601-1XX	Kit, Tool, Special

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, N.J. 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Availability of documents. Applicable (see 6.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any other document.

(b) This specification - This specification shall have precedence over all applicable subsidiary documents (see 6.6).

(c) Referenced documents - Any document referenced in this specification shall have precedence over all applicable documents referenced therein,

3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable except as follows (see 6.6).

3.1.1 Selection of materials, parts, and processes. Applicable.

3.1.1.1 Approval of nonstandard materials, parts, and processes. Applicable.

3.1.2 Flammable materials. Applicable.

3.1.3 Cables and connectors. Not applicable.

3.1.4 Wire. Not applicable.

3.1.5 Rotary switches. Not applicable.

3.1.6 Finishes. Not applicable.

3.1.7 Marking and identification. Not applicable.

3.1.8 Soldering. Not applicable.

3.1.9 Welding. Applicable.

3.1.10 Printed wiring. Not applicable.

3.2 Design and Construction. Applicable except as follows.

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3.2.1 Design features. Applicable except as follows.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Not applicable.

3.2.1.2.1 Other commercial components. Not applicable.

3.2.1.3 Circuit grounds. Not applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Not applicable.

3.2.1.8 Forced-liquid cooling. Not applicable.

3.2.1.9 Space and weight. The maximum dimensions of the STK box shall be 24 inches by 9 inches by 10 inches. The maximum weight of the STK (including the tools) shall be 40 pounds.

3.2.1.10 Connectors. Not applicable.

3.2.1.11 Secondary electrical power. Not applicable.

3.2.1.12 Interface. The tools of Table I shall interface with the applicable test stations and WRAs.

3.2.2 Utilities. Applicable.

3.2.2.1 Electrical power. Applicable. The electrical items of the STK shall be capable of operating from 115 volt, 60 Hertz, Type I power.

3.2.2.2 Clean Air. Not applicable.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Not applicable.

3.2.2.5 Helium. Not applicable.

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3.2.2.6 Nitrogen. Not applicable.

3.2.2.7 Hydraulic fluid. Not applicable.

3.2.3 Service conditions.

3.2.3.1 Electrical. The electrical power required for operation of STK items shall be 115 volts ac, 60 Hertz (Hz) single phase. Maximum current requirements shall be 5 amperes.

3.2.3.2 Envirnomental. Applicable, except that the commercial items shall conform to the equipment manufacturer's specifications.

3.2.5.2.1 Organizational level. Not applicable.

3.2.3.2.2 Intermediate Ievel. Applicable.

3.2.4 Maintainability. Applicable.

3.2.4.1 Mean time to repair. Not applicable.

3.2.4.2 Test points. Not applicable.

3.2.4.3 Shop replaceable assemblies (SRA). Not applicable.

3.2.4.4 Self test capabilities. Not applicable.

3.2.5 Reliability. Not applicable.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Not applicable.

3.2.6 Electromagnetic interference (EMI). Not applicable.

3.2.7 Utilities protection. Not applicable.

3.2.7.1 Liquid cooling protection. Not applicable.

3.2.7.2 Clean air protection, Not applicable.

3.2.7.3 Electrical power protection. Electrical items in the STK shall operate with 3-wire grounding plugs.

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3.2.8 Calibration cycle. Not applicable.

3.3 Performance.

3.3.1 General. The STK shall consist of a tool box (Kennedy Part No. K-24 or equivalent) containing the items listed in Table I for maintaining certain weapon replaceable assembles (WRAs) used in AN/AWG-9, and portions of AN/AWM-23(V) Test Set.

3.3.2 Organizational level functional requirements. Not applicable.

3.3.3 Intermediate level functional requirements. Applicable.

3.3.4 Warm up time. Not applicable.

3.4 Detailed functional requirements. The tools listed in Table I shall be used in repairing and maintaining the WRAs tested at the various test stations specified, and for repairing and maintaining the AN/AWM-23(V) Test Set.

3.5 Workmanship. Applicable.

3.6 Elements. See Table I.

4. Quality assurance provisions.

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

4.2.5 Acceptance test plan. Applicable. Table II shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance. Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

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TABLE 1
LIST OF SPECIAL TOOL KIT ITEMS REQUIRED FOR
MAINTAINING AN/AWG-9 WRAs AND AN/AWM-23(V)

ITEM	NOMENCLATURE	MANUFACTURER AND PART NO. OR EQUIVALENT	AN/AWG-9 WRAs MAINTAINED AT THE APPLICABLE TEST STATIONS				AN/AWM-23(V) ITEMS MAINTAINED
			LOW FREQUENCY TEST STATION	COMPUTER TEST STATION	RADIO FREQUENCY TEST STATION	CONTROLS-DISPLAYS-DOPPLER FILTER TEST STATION	
1	Kit, Maintenance - Removable Contact, GSC Crimping	Buchanan Electrical Products 615673	481031 481081 481101 481120 481601 481610 481730	481451 481452 481710 481720 481461 481462	481001 481010 481011 481013 481014 481015 481022 481039 481083	481042 481501 481505 481541 481560 481580 481590	493502
2	Tool, Wire Unwrapping, LH Manual, 20/26 AWG	Gardner/ Denver Company A31478	481081 481610	481720 481451 481710	481010 481083 481039	481042 481541 481580	----
3	Tool, Wire Unwrapping, LH Tool, Electric	Gardner /Denver Company 500315	481081 481610 481730	481720 481710 481461 481462	481039 481010 481083	481042 481541 481580	----
4.1	Nose Assy, STD	500108	481081	481720	481039	481042	
4.1.1	Bit (18 AWG)	504908	481610	481710	481010	481541	----
4.1.2	Sleeve (18 AWG)	18640		481461	481083	481580	----
4.1.3	Bit (20 AWG)	26495					
4.1.4	Sleeve (20 AWG)	26245					----
4.2	Nose Assy, Backforce (24 AWG)	500318	481081	481720	481039	481042	
4.2.1	Bit (24 AWG)	26263	481610	481710	481010	481541	----
4.2.2	Sleeve (24 AWG)	18840		481461	481083	481580	
4.3	Nose Assy, Backforce (30 AWG)	507395					
4.3.1	Bit (30 AWG)	507063	481730	481462	----	----	----
4.3.2	Sleeve (30 AWG)	507100					
5	Hand-Tool, Termi-Point	AMP INC 69525-1	----	----	----	----	493150 493250 493350 493450
5.1	Mandrel, Termi-Point	69411-4					
6	Extractor, Termi-Point	AMP INC. 69357-1 or-3					
7	Extractor, Contact, Termi-Point	AMP INC. 465199-1					

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TABLE I (Cont'd.)

ITEM	NOMENCLATURE	MANUFACTURER AND PART NO OR EQUIVALENT	AN/AWG-9 WRAs MAINTAINED AT THE APPLICABLE TEST STATIONS				AN/AWM-23(V) ITEMS MAINTAINED
			LOW FREQUENCY TEST STATION	COMPUTER TEST STATION	RADIO FREQUENCY TEST STATION	CONTROLS-DISPLAYS-DOPPLER FILTER TEST STATION	
8	Extractor, P/C Card	Flotron, Inc. 6126-413	481081 481730 481610	481720 481710 481461	481010 481039 481085	481042	----
9	Extractor, P/C Card (Adjustable)	Flotron, Inc 6126-452	481081 ----	481720 ----	481010 ----	481042 481505 481541 481580	----
10	Extractor, P/C Card (Computer)	Control Data Corp. 16805400 or 59267000	----	481451 481452	----	----	----
11	Extractor, P/C Card (Radar)	HAC 3229133	481610	----	481001	----	----
12	Stripper, Wire, Thermal	Kinetics Corp TW-1	----	----	481039 481590	----	----
13	Tool, Removal, IR Dome	HAC 3229106	481101	----	----	----	----
14	Kit, Boresight Assembly	HAC 3111320	----	----	----	481590	----
15	Wrench, Allen, Special	HAC 3229107	----	----	481011	----	----
16	Tool, Cap Securing	HAC 10191465 (Newport Beach)	----	----	----	----	493502
17	Extractor, Lead	Tech Devices CO 91-1-60	----	----	481010	----	----
18	Tool, Wire, Unwrapping, Manual	Gardner/Denver Company 505244	----	----	----	----	493400
19	Wrench, Bristol	Bristol Co S048-4	----	----	----	481560	----
20	Card Puller	HAC 3282248	481001	----	----	----	----
21	Socket, Hex Bit	HAC 3282246	481011	----	----	----	----
22	Card Puller	HAC 3282245	----	----	----	----	493150 493250 493350
23	Tool, Wire, Unwrapping, Ld. Manua.	Gardner/Denver Company 505084	481730	481462	----	----	----

See 6.3 for AN/AWG-9 WRA and AN/AWM-23(V) Nomenclature

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TABLE II

ACCEPTANCE TEST PLAN FOR SPECIAL
TOOL KIT
TK-222( )/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate that the material, workmanship and mechanical interchangeability requirements of 3.1, 3.5 and 3.2.1.4 respectively have been met.

2.0 TEST CONDITIONS (None applicable)

3.0 TESTS. Inspect each item of the special tool kit for conformance to drawing 493601-1XX.

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4.3 Tests.

4.3.1 Electrical and electronic items - operational run-in test. Not applicable.

4.3.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable,

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The STK is intended for use at the intermediate level for testing and maintaining various AN/AWG-9 WRAs and AN/AWM-23(V) units.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.

6.4 Nomenclature cross reference. The nomenclature of the AN/AWG-9 WRAs and the AN/AWM-23(V) items specified in Table I is listed in Table III for reference only.

6.5 Liquid cooler. Not applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-K-81779(AS) are the same as MIL-T-81734(AS) the word “applicable” is used in this specification. When the term “not applicable” is used in this specification, it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met (see 6.6 of MIL-T-81734(AS)).

6.7 Abbreviations and acronyms. Applicable. In addition, the following shall apply:

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TABLE III

AN/AWG-9 WRA AND AN/AWM-23(V) NOMENCLATURE

Item	Unit No.	Nomenclature
1	481001	Oscillator, Radio Frequency O-1633( )/AWG-9
2	481010	Synchronizer-Transmitter SN-487( )/AWG-9
3	481011	Transmitter, Radar T-1224( )/AWG-9
4	481013	Power Supply PP-6770( )/AWG-9
5	481014	Power Supply PP-6771( )/AWG-9
6	481015	Power Supply PP-6772( )/AWG-9
7	481022	Receiver, Radar R-1785( )/AWG-9
8	481031	Antenna AS-2764( )/AWG-9
9	481039	Converter, Signal Data CV-2950( )/AWG-9 and CV-3225( )/AWG-9
10	481042	Filter-Multiplexer F-1331( )/AWG-9
11	481081	Control, Antenna-Test C-9038( )/AWG-9 and C-10036( )/AWG-9
12	481083	Converter, Signal Data CV-2946( )/AWG-9 CV-3224( )/AWG-9 and CV-3366( )/AWG-9
13	481101	Receiver, Infrared R-1786( )/AWG-9
14	481120	Processor, Data CV-2947( )/AWG-9
15	481451	Computer, Digital CP-1159( )/AWG-9
16	481452	Memory Unit MU-588( )/AWG-9
17	481461	Processor, Data CV-3151( )/AWG-9 and CV-3165( )/AWG-9

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TABLE III (Continued)

Item	Unit No.	Nomenclature
18	481462	Power Supply-Storer PP-6776( )/AWG-9
19	481501	Control, Sensor C-9035( )/AWG-9
20	481505	Control, Computer C-9036( )/AWG-9
21	481541	Display, Detail Data IP-1111( )/AWG-9, IP-1173( )/AWG-9 and IP-1185( )/AWG-9
22	481560	Control, Sensor C-9037( )/AWG-9
23	481580	Display, Tactical IP-1110( )/AWG-9
24	481590	Recorder, Mission Data KF-21( ) and KF-22( )
25	481601	Power Supply PP-6773( )/AWG-9
26	481610	Power Supply PP-6774( )/AWG-9
27	481710	Converter, Signal Data CV-3152( )/AWG-9
28	481720	Control, Logic- Timing C-9506( )/AWG-9
29	481730	Power Supply PP-7010( )/AWG-9 and PP-7042( )/AWG-9
30	481990	Rack, Electrical Equipment MT-4433( )/AWG-9
31	493150	Test Station, Radio Frequency OJ-213( )/AWM-23(V)
32	493250	Test Station, Controls-Displays-Doppler Filter OJ-214( )/AWM-23(V)
33	493350	Test Station, Low Frequency OJ-215( )/AWM-23(V) and OJ-379( )/AWM-23(V)
34	493400	Test Station, Computer OJ-216( )/AWM-23(V)
35	493457	Test Station, Module OJ-217( )/AWM-23(V) and OJ-378( )/AWM-23(V)

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TABLE III (Continued)

Item	Unit No.	Nomenclature
36	493502	Test Set, Missile Auxiliaries TS-3479( )/ AWM-23(V)
37	493601	Kit, Tool, Special TK-222( )/AWM-23(V)

HAC Hughes Aircraft Company
STK Special Tool Kit

6.8 Convenience outlet power. Not applicable.

6.9 Changes from previous issue are not marked.

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